

Scientific Programming – Group Project

Title

Tagesschau Mining - Data Acquisition

Background

Data science is an interdisciplinary field concerned with scientific methods, processes, and systems used to extract knowledge or insights from data in various forms, either structured or unstructured, and includes areas such as machine learning, computational statistics, data mining, predictive analytics, and other emerging related fields of study. The field involves preparing data for analysis, formulating data science problems, analyzing data, developing data-driven solutions, and presenting results to inform high-level decisions in a wide range of application domains. An interesting example of such a data science project is SpiegelMining, which was presented at the chaos communication congress (33c3).

In a similar approach, we want to uncover the inner workings of the public broadcaster's news website www.tagesschau.de using the tools of data science. Three parallel projects will each focus on a cornerstone of data science: data acquisition, data assembly, and data mining.

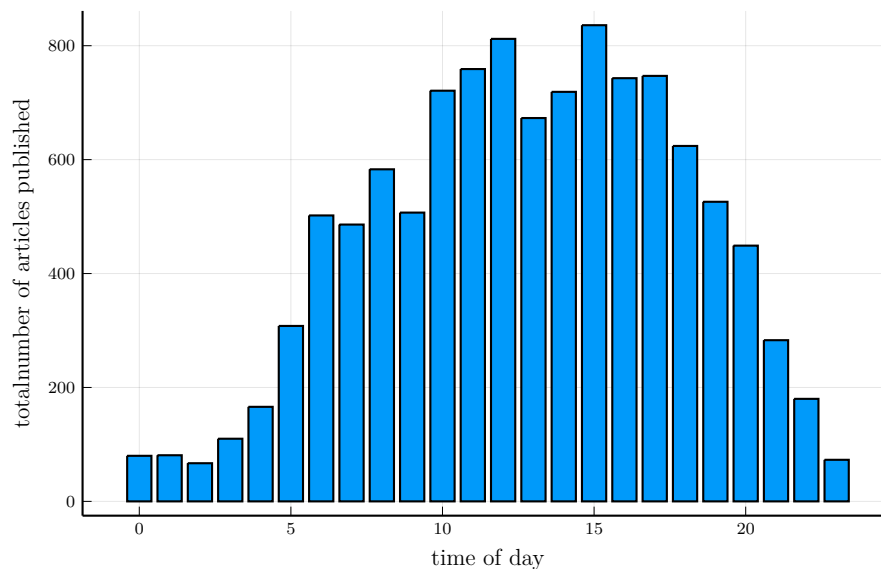


Figure 1: From the Tagesschau data, one can find out when articles are published. The two dips between 9 and 10 a.m. and at noon could be due to a regular editorial meeting and lunch break.

Project Tasks

`tagesschau.de` offers a public interface <https://www.tagesschau.de/api2/homepage/>, where the contents of the website are provided in the JavaScript Object Notation (JSON) format. This format uses human-readable text to store and transmit data objects consisting of attribute–value pairs and arrays.

The objective of this project is to develop a data acquisition framework for the `tagesschau.de`:

1. Use the public interface <https://www.tagesschau.de/api2/homepage/> to acquire data (json format) from the `tagesschau.de` website in 5 min intervals for at least a week.
2. Store the acquired data on disk.
3. Name each json-file by its acquisition date time.
4. Each file is about 1.6 MB in size, so roughly 500 MB of data will be incoming per day. Use compression methods keep the storage requirements minimal.
5. Write methods to log the process, handle exceptions and monitor the progress.
6. Use Julia as much as possible for all these tasks.

General Task

In addition to the project specific task you are expected to take a dive into proper software development. In particular we want you to:

1. Create a road map, which sketches the tasks to be done and milestones to be reached.
2. Work as a team. Make a kick off meeting where you decide jointly how the workload can be partitioned and how the common interface should look like.
3. Use Gitlab to host your project and collaborate. To this end we want you to create and resolve at least one issue and create / merge at least one pull request.
4. Document your project. The `Readme.md` is a good place to introduce your project and how to use it, whereas the methods, which you consider public can be assigned docstrings.
5. Write unit tests for your core methods to ensure they work as intended.
6. Prepare a 5 minutes talk to present your project to the other course participants at the end of the project.

We will at the end of the project go through this general task list and check, which points you have taken into account for the grading.

Project schedule

The schedule of the project looks as follows. Week zero is the 16.6.2023. Week 4 is the 14.7.2023.

